# ONUR SEL

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## SUMMARY

Fourth-year Computer Engineering student (expected graduation 2027) seeking a long-term software engineering internship for the Spring 2027 semester. Hands-on experience in embedded systems and computer vision (research internship on a robot-mounted tracking system), machine learning, backend and database work, and shipping production apps to real users. I like measuring systems and finding root causes, and I work well both in a team (hackathon, research group) and on my own, from design through release.

## EDUCATION

**B.Sc. in Computer Engineering (English-medium)** 2022 – 2027 (expected)
*Abdullah Gül University, Kayseri*

- Fourth-year student. Coursework: Data Structures & Algorithms, Object-Oriented Programming & Design Patterns, Operating Systems, Computer Networks, Database Management Systems, Software Development Life Cycle & Agile.

## EXPERIENCE

**Research Intern – Robot-Mounted Visual Tracking System** Jul 2026 – Aug 2026
*Abdullah Gül University, Kayseri (hybrid) – under Asst. Prof. Khaled Hejja*

- Co-authored a journal manuscript on a camera-based pan-tilt tracking system mounted on a quadruped robot; it has passed the initial check and is under peer review at the Journal of Materials and Mechatronics: A (DergiPark). Prepared a second manuscript as well.
- Audited a split Raspberry Pi 5 (OpenCV vision) / Arduino (servo control) architecture at code level and corrected inconsistencies between the documentation and the implementation.
- Measured end-to-end latency on hardware (~46 ms); built a digital twin to validate experiments in simulation.
- Tested neural-network-based controllers, found bugs that prevented target lock, and benchmarked them against a classical controller.

## PROJECTS

**Selware – Independent Mobile App Studio** 2025 – Present
*Flutter · Dart · Firebase · RevenueCat · AdMob | selwareapps.com*

- Design, develop, publish and maintain my own apps end to end, including store listings, privacy policies, in-app purchases and ads, and policy compliance.
- 4 apps live on both Google Play and the App Store: Finansel (personal budget tracking and analytics), Ezan Vakitleri (prayer times), Adaş (name meanings and origins) and Pip & Pop (two-player games).

**Study Room – Real-Time Shared Study Room** 2026
*Java 21 · Spring Boot · WebSocket (STOMP) · Redis · PostgreSQL · React · TypeScript · Docker | Live demo · GitHub*

- Account-free shared Pomodoro room in the browser: server-validated timer in Redis, live sync over WebSocket, anonymous JWT identity, and authorization where only the room's creator can change the lengths; 44 tests, live on Render + Neon.

**ExcuseMan – Pixel Game and Leaderboard Backend** 2026
*Flutter · Dart · Java 21 · Spring Boot · PostgreSQL · Docker · Testcontainers*

- 2D pixel platformer/fighter (9 worlds, 86 levels, endless mode) with my own Spring Boot world leaderboard: server-validated runs, per-run HMAC anti-cheat, 143 tests, CI; live on Render + Neon. Not on the stores yet.

**Gams N' Roses – Optimization Game** 2026
*Team project · 1st place, Gamification for Optimization Hackathon | GitHub*

- Game where players pick the best city for a concert venue, combining optimization, location analysis and gamification.

**AI-Generated Image Detection** 2026
*Python · OpenCV · scikit-learn · NumPy · FFT · LBP | GitHub*

- Explainable, lightweight feature-extraction pipeline that separates generative-AI images from real photos.

**Secure Email System** 2025
*Python · AES / RSA / SHA-256 · Socket Programming | GitHub*

- Encrypted email client/server with hybrid AES + RSA encryption and SHA-256 integrity checks over sockets.

## TECHNICAL SKILLS

**Languages:** Dart, Python, Java, C, SQL, JavaScript, TypeScript
**Mobile & Backend:** Flutter, Firebase, Spring Boot, PostgreSQL, Redis, WebSocket, Docker, Node.js, REST APIs, Socket Programming
**Vision & ML:** OpenCV, scikit-learn, NumPy
**Embedded & Tools:** Arduino, Raspberry Pi, Git, GitHub, Linux
**Spoken Languages:** Turkish (native), English (B2, upper-intermediate)